

6.DETERMINATION OF RIGIDITY MODULUS OF A WIRE MATERIAL **- TORSIONAL PENDULUM**

Aim:

To determine the rigidity modulus (η) of the material of the given wire using Torsional pendulum

Apparatus:

Torsional pendulum, thin wire, stops watch, screw gauge, Vernier Calipers, Scale.

Formula:

$$\eta = \frac{4\pi m R^2}{a^4} \left(\frac{l}{T^2} \right)$$

Where, m = Mass of the disc

R = Radius of the disc

T = Time period of oscillation

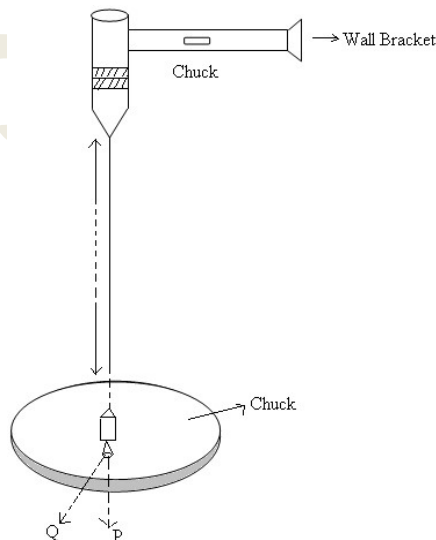
a = radius of the wire

l = length of the wire between chucks

η = Rigidity modulus of the wire

Description:

The Torsional pendulum consists of a uniform circular metal disc of about 8 to 10 cm diameter with 1 (or) 2 cm thickness suspended by a wire at the centre of the disc as shown in fig. the lower end of the wire is gripped into the chuck at the centre of the disc and the upper end is gripped in to another chuck which is fixed to a wall bracket.



Procedure:

The circular metal disc is suspended as shown in fig. The length of the wire is adjusted at 90 cm. A small mark is made on the circular disc is noted the number of oscillations.

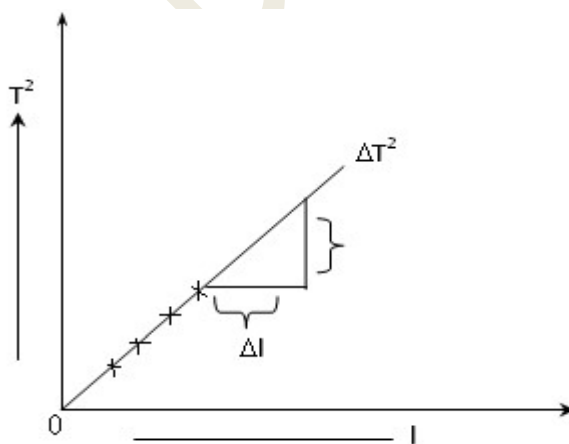
The disc is set to oscillate slowly turning the disc through a small angle.

When the disc is oscillating the time taken for 20 oscillations is noted with the stop clock and the observations are tabulated as Trail-1. The procedure is repeated for the same length and time for 20 oscillations is noted as Trail – 2.

The mean value of trail – I and trial – II is calculated the time period (T) i.e. the time taken for one oscillation is calculated. The experiment is repeated by decreasing the length of the wire in steps of 10 cm and the results are calculated (or) tabulated.

The radius of the wire 'r' is found with screw gauge. The radius and the mass of the disc are found with vernier calipers and rough balance.

The mean value of $\frac{l}{T^2}$ is substituted in above formula and experimental values of η is to be found: A Graph is drawn with 'l' on the x-axis and T^2 on the y - axis $\frac{l}{T^2}$ is calculated from the graph. The graphical value of ' η ' is said to be found by substituting $\frac{l}{T^2}$ is the above equation.



Precautions:

1. Oscillations are observed without parallax error
2. The movement of the disc should not be found

Table – 1

S. No.	Length of the wire 'l' between chucks (cm)	Time taken for 20 oscillations (s)			Time period $T = \frac{x}{20}$ (s)	T^2 (s ²)	$\left(\frac{l}{T^2}\right) \text{cm/sec}^2$
		Trail I	Trail II	Average (x)			
1.							
2.							
3.							
4.							

Average $\left(\frac{l}{T^2}\right) =$

Table – 2: Diameter of the disc using Vernier calipers

Least count (L.C.) = 0.01 cm

S. No.	M.S.R a (cm)	VC	V.CXL.C. (b) cm	Total Reading (a+b) cm
1.				
2				
3.				

Average radius (R) = _____ cm

Table – 3: Diameter of the wire using screw gauge:

Least count (L.C.) = 0.01 mm

S.No	P.S.R a (mm)	Head scale coincidence		Fraction $b=n \times \text{L.C}$ mm	Diameter (a+b) mm	Radius $r = \frac{\text{diameter}}{2}$ mm
		Observed	Corrected (n)			

Average radius (r) = _____ mm

= _____ cm

Observations:

$$\text{Experimental value } \eta = \frac{4\pi MR^2}{r^4} \left[\frac{l}{T^2} \right] = \text{_____ dyne/cm}^2$$

$$\text{Graphical value } \eta = \frac{4\pi MR^2}{r^4} \left[\frac{\Delta l}{\Delta T^2} \right] = \text{_____ dyne/cm}^2$$

Calculations:

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Result:

1. The experimental value of η = _____ dyne/cm²
2. The graphical value of η = _____ dyne / cm²

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